

M.E EDUCATION CENTRE

32/B, Mehmoodabad-6, Karachi; Ph: 35891008

CHEMISTRY: 12 - HALF-YEARLY EXAM: 2025-26

SECTION "A" (SHORT QUESTIONS / ANSWERS)

NOTE: Attempt any FIVE questions from Inorganic and FIVE questions from Organic Chemistry. (10x3=30)

INORGANIC CHEMISTRY

- (i) Give reasons for the following:
(a) Why do transition elements show variable oxidation state?
(b) Why Cu^{+2} ion is blue but Zn^{+2} ion is colourless?
- (ii) Write down the balanced chemical equation for the following reactions:
(a) Reaction of permanganate with oxalic acid
(b) Reaction of Iron with Chlorine
- (iii) Write down the names of steel types and applications?
- (iv) Write the IUPAC name of the following:
(a) $[\text{Cr}(\text{en})_3](\text{NO}_3)_3$ (b) $[\text{Pt}(\text{NH}_3)_2\text{Cl}]_2^{2+}$ (c) $\text{Na}_3[\text{Co}(\text{C}_2\text{O}_4)_3]$
- (v) What do you mean by Chelates and Chelating agent? Give an Example
- (vi) What is flame test? Mention the color of flame of alkali metals.
- (vii) Write down the unique behavior of Beryllium in group II-A.

Steel
tools
stainers

ORGANIC CHEMISTRY

- (i) Write the uses and properties of Lignite and Anthracite?
- (ii) Explain the partial and total synthesis?
- (iii) Explain the reforming of petroleum?
- (iv) Give methods of preparation of Alkenes?
- (v) Give hydration, epoxidation and ozonolysis in ethene?
- (vi) Give acidity of terminal alkynes with two examples?
- (vii) What do you mean by Chiral centre? Give an example

SECTION "B" (LONG QUESTIONS / ANSWERS)

NOTE: Attempt any TWO questions from Inorganic and TWO questions from Organic Chemistry. (4x5=20)

INORGANIC CHEMISTRY

- (i) Describe the manufacturing of H_2SO_4 by the Contact Process.
- (ii) Describe the manufacture of Caustic Soda by the Castner-Kellner Cell
- (iii) Describe the various steps involved in the extraction of 99.99% pure copper from its chalcopryite ore.

ORGANIC CHEMISTRY

- (i) Explain the molecular orbital treatment of benzene
- (ii) What are organometallic compounds? How is Grignard reagent prepared? Write the reactions of CH_3MgI with Water, Alcohol and Ester.
- (iii) Write chemical equations of the following:
* Reaction of ethanol with acetic acid in the presence of H_2SO_4 .
* Reaction of phenol with concentrated HNO_3 .
* Reaction of alcohol with concentrated H_2SO_4 .
* Reaction of ethyle bromide with alcoholic KOH
* Reaction of benzene with fuming sulphuric acid
- (iv) What is meant by a nucleophile? Give the mechanism of the following reactions:
(a) S_N^1 reaction between a 3° alkyl halide and NaCN .
(b) S_N^2 reaction between a 1° alkyl halide and NaOH .

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MATHEMATICS: XII

HALF YEARLY EXAM: 2026

Timings: 2.0 Hr.

Total Marks: 50

Note: (1) Attempt EIGHT (8) Questions from Section: A (3 Marks each) including Q:1 and 2 are compulsory (4 marks each). (2) Attempt FOUR (4) Questions from Section: B (6 Marks each).

SECTION: A

- ✓ 1. Find the derivative by ab-initio: $f(x) = \sqrt{\tan x}$ (Compulsory)
- ✓ 2. A particle moves along the curve $x = 2t^2$ and $y = 4t$. Find the components of velocity and acceleration at $t = 2$ in the direction of $2\hat{i} - \hat{j}$. (Compulsory)
3. Test continuity and discontinuity of the following function: $f(x) = \begin{cases} 2+x, & \text{when } x < 3 \\ 5-2x, & \text{when } x \geq 3 \end{cases}$ at $x = 3$
4. Evaluate the limit of anyone: (i) $\lim_{x \rightarrow e} \frac{\ln x - 1}{x - e}$ (ii) $\lim_{x \rightarrow 1} \frac{\frac{1}{\sqrt{x}} - 1}{1 - x}$
- ✓ 5. Differentiate: $y = \cos^2 x$ w.r.t. $\sin^2 x$.
- ✓ 6. If $y = \sqrt{\tan x} + \sqrt{\tan x} + \sqrt{\tan x} + \dots + \infty$, prove that $(2y - 1) \frac{dy}{dx} = \sec^2 x$. *Sec² x*
- ✓ 7. If $y = \tan^{-1} \left(2 \tan \frac{x}{2} \right)$, then prove that $\frac{dy}{dx} = 4 \left(\frac{1+y^2}{4+x^2} \right)$ *1 / (2 tan(x/2))*
- ✓ 8. Show that $f(x) = \tan^2 x$ is decreasing at $x = \frac{3\pi}{4}$. *t ≠ 1, t > 0*
9. For what value of t , the vector function $\vec{r}(t) = \ln(t+3)\hat{i} + \frac{1}{t-1}\hat{j} + \frac{1}{e^t}\hat{k}$ is continuous.
10. Find the area above the x-axis under the following curve, between the given ordinates. $y = 6 \sin^2 x$, $x = 0, x = \frac{\pi}{3}$
11. The x-intercept of a line is the reciprocal of its y-intercept and line pass through $(2, -1)$. Find its equation.
- ✓ 12. The gradient of one of the lines $ax^2 + 2hxy + by^2 = 0$ is twice that of the other. Show that $8h^2 = 9ab$.

$$\frac{-m + \sqrt{h^2 - ab}}{b} \quad 2 \left(\frac{-m - \sqrt{h^2 - ab}}{b} \right)$$

SECTION: B

- ✓ 13. (i) Obtain the first three terms of the Maclaurin's series for $f(x) = \cos x$.
 (ii) Find the first four terms of the Taylor's series for the function $f(x) = \sin x$ at $a = \frac{\pi}{4}$.
- ✓ 14. Find the derivatives of any two: (i) $y = \tan^{-1} \left(\frac{\sin 2x}{1 + \cos 2x} \right)$ (ii) $y = \ln \left[\tan \left(\frac{x}{2} + \frac{\pi}{4} \right) \right]$ (iii) $x^y \cdot y^x = 1$
- ✓ 15. Evaluate any two: (i) $\int \frac{1}{x + \sqrt{x}} dx$ (ii) $\int \frac{\sin^3 x}{\sqrt{\cos x}} dx$ (iii) $\int \frac{x^5 dx}{\sqrt{x^2 - 9}}$
16. Evaluate any two: (i) $\int \sec^3 x dx$ (ii) $\int \frac{(5x+6)dx}{(x+3)(x-2)^2}$ (iii) $\int x \cot^{-1} x dx$
17. Compute by using basic properties of any two: (i) $\int_0^{\frac{\pi}{2}} \cos^3 x dx$ (ii) $\int_{\frac{\pi^2}{4}}^{\frac{\pi^2}{36}} \frac{\sin \sqrt{x}}{\sqrt{x}} dx$ (iii) $\int_{\frac{\pi}{2}}^{\frac{\pi}{4}} \tan^2 x dx$
18. If A (2,5), B (3,7) and C (0,8) are the vertices of a triangle then find the equation of (a) median through A (b) right bisector of side AC

GOOD LUCK

Farhan Yousuf Khan

Head of the Institutions: Farhan Yousuf Khan



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2024-25

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IX-X-XI-XII
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PAKISTAN STUDIES: 12 – HALF-YEARLY EXAM: 2025-26

SECTION "A" (SHORT ANSWERING QUESTIONS)

Note: Attempt only SEVEN Questions, including Q.1, is compulsory.

- Q.1: Describe the social conditions of south Asia before the advent of Islam. (06)
۱۔ اسلام سے قبل جنوبی ایشیا کے معاشرتی حالات بیان کیجئے۔
- Q.2: Write any four Islamic provinces of 1973 constitution. (04)
۲۔ 1973 کے آئین کی چار اسلامی دفتوں تحریر کریں۔
- Q.3: Write the importance of the Northern Mountains of Pakistan. (04)
۳۔ پاکستان کے شمالی پہاڑوں کی اہمیت بیان کریں۔
- Q.4: What is the Indus Basin Water Treaty? (04)
۴۔ سندھ طاس معاہدہ کیا ہے؟
- Q.5: Describe four points of Lucknow Pact 1916. (04)
۵۔ لکھنؤ معاہدہ کے چار نقاط تحریر کریں۔
- Q.6: Describe the objectives of the Khilafat Movement. (04)
۶۔ خلافت تحریک کے مقاصد بیان کریں۔
- Q.7: Write any four points from the fourteen points of Quid-e-Azam. (04)
۷۔ قائد اعظم کے چودہ نقطوں میں سے کوئی چار نقاط تحریر کریں۔
- Q.8: Write any two problems faced by Pakistan after its creation. (04)
۸۔ پاکستان کی ابتدائی چار مشکلات تحریر کریں۔
- Q.9: What is the importance of Objectives Resolution in the Constitution making of Pakistan? (04)
۹۔ قرارداد مقاصد کی اہمیت بیان کریں۔
- Q.10: State the Social impact of Islam on South Asia. (04)
۱۰۔ جنوبی ایشیا پر اسلام کے معاشرتی اثرات بیان کریں۔

SECTION "B" (DETAILED ANSWERING QUESTIONS)

Note: Attempt only TWO Questions.

- Q.11: What do you mean by ideology of Pakistan? Describe in the light of Quaid-e-Azam's statements. (10)
نظریہ پاکستان سے کیا مراد ہے؟ قائد اعظم کے ارشادات کی روشنی میں واضح کریں۔
- Q.12: Describe the importance of geographical location of Pakistan. (10)
پاکستان کے جغرافیائی محل وقوع کی اہمیت بیان کریں۔
- Q.13: Describe the important political events that took place during the period between the Pakistan Resolution 1940 and establishment of Pakistan 1947. (10)
1940 سے 1947 کے درمیان ہونے والے اہم سیاسی واقعات بیان کریں۔
- Q.14: State the services of Quaid-e-Azam Muhammad Ali Jinnah in the creation of Pakistan. (10)
قیام پاکستان کیلئے قائد اعظم کی خدمات بیان کیجئے۔

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ENGLISH: 12 - HALF-YEARLY EXAM: 2025-26

SECTION "A" (SHORT-ANSWER-QUESTIONS)

Q.1: Attempt Ten (10) part-questions from this section all together, taking two-part questions from each sub-section. Answer should not exceed Six sentences. All questions carry equal marks. (40)

SUB-SECTION I (INTERMEDIATE ENGLISH BOOK-II)

1. (i) How does the Miller deceive Little Hans?
2. (ii) Why does Martin Luther King emphasize that the struggle for equality must be conducted with dignity and discipline?
3. (iii) How long did the panic last and how was the order restored?
4. (iv) How did the panic start in the movie theatre / cinema on the High Street?
5. "OR" What message does Martin Luther King convey to his listeners through his speech?

SUB-SECTION II (SELECTION FROM ENGLISH VERSES: PART-II)

6. (v) How does Kiplin suggest one should deal with both success and failure?
7. (vi) What role does curiosity play in the poem, and how does it shape the speaker's experience of the world?
8. (vii) How does Ulysses motivate his companion to sail once again? And why?
9. (viii) Give critical review of the poem "Ulysis"?
10. "OR" Write fourth stage according to the poem "Seven Ages of Man"

SUB-SECTION III (NOVEL)

11. (ix) What is the Central Theme of the novel "Great Expectations"?
12. (x) Is Pip happy about becoming a blacksmith after visiting Miss Havisham's house?
13. (xi) What is unusual about Miss Havisham's appearance lifestyle, and how do these reflect her past?
14. (xii) Does Pip tell the truth about his visit to Miss Havisham's house? Does Pip reveal the truth to someone later? "OR" How does Estella treat Pip when they first meet, and in what ways does the encounter influence his self-image?

SUB-SECTION IV (GRAMMAR)

15. (xiii) Spot the errors and do the correction of verb, preposition and adjective in the following sentences.
 - (a) She have been busy in her project for many days. *has*
 - (b) He always lie to others. *lies*
 - (c) My income is little than that of my elder brother. *less*
 - (d) They did not brought up their children well. *bring*
 - (e) The child did not fell off the staircase every day. *fall*
16. (xiv) Write meanings only four of the following words.

(a) Inquisitiveness	(b) Conviction	(c) Cynical	(d) Captivity
(e) Despair	(f) Destiny	(g) Discrimination	(h) Pledge
17. (xv) Change the narration:
 - (a) The manager said to his assistant, "Get out of my office and do not come back".
 - (b) Madam said, "More haste less speed."
 - (c) My friend said to me, "I will not pay back your money now."
 - (d) I said to her, "What did you say to me?"
18. (xvi) Change into active voice:
 - (a) The parcel was delivered by the courier.
 - (b) Was the problem solved by the engineer?
 - (c) Who was the award given to by the jury?
 - (d) How was the evidence destroyed by the criminals?

*award was given to asim by Jury
Jury gave the award to asim.
asim was the award winner*

SECTION "B" (DETAILED-ANSWER QUESTIONS)

Q.2: Write a detailed character sketch of one of the followings: (i) Pip (ii) Estella

(10)

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PHYSICS: XII

HALF YEARLY EXAM: 2026

Timings: 2.0 Hr.

Total Marks: 50

Note: Attempt SIX (6) Questions from Section: A (4 Marks each) and TWO (2) Questions from Section: B (7 + 6 = 13 Marks each).

SECTION: A

1. A heat engine works at the rate of 500kW. The efficiency of the engine is 30%. Calculate the loss of heat per hour.
2. State first law of thermodynamics. Describe isothermal process.
3. Suppose we have a toroid with 1000 turns and a current of 3 amperes flowing through it. The toroid has a circular cross-section with a radius of 0.1 meters. Calculate the magnetic flux density inside the toroid.
 $4\pi \times 10^{-7}$
4. A sample of a gas contains 3.0×10^{24} atoms. Calculate the volume of the gas at a temperature of 300K and a pressure of 120K Pa.
5. How energy is stored in a current carrying inductor. Derive the formula for energy produced in self-induction.
6. How can a galvanometer be converted into an ammeter? Derive the relevant expression.
7. The current in a coil of 500 turns is changed from 0 to 5.43 amps. Thereby producing a magnetic flux of 8.52×10^{-4} w. What is the Self-Inductance of the coil?
8. The thickness of a metal plate is 0.35 inches. It's drilled to have a hole of radius 0.08 inches on the plate. If the shear strength is 4×10^4 lbs/in², determine the force needed to make that hole.
9. In an RL circuit, the resistance (R) is 30 ohms, and the inductance (L) is 0.2 H. Calculate the total impedance at a frequency of 60 Hz?
10. Define and explain Curies' law.
11. What is the flux density at a distance of 0.1m in air from a long straight conductor carrying a current of 6.5 A? Calculate the force per meter on a similar parallel conductor at a distance of 0.1m from the first carrying a current of 3A. ($\mu_0 = 4\pi \times 10^{-7}$ T.m.A⁻¹)
 $B = \frac{\mu_0 I}{2\pi r}$ $F = B \cdot I \cdot l$
12. Prove that the difference of two molar specific heat capacities of a gas is equal to universal gas constant.

SECTION: B

13. (a) Describe the construction and working of Moving coil galvanometer. Also, show that the deflection produced in the coil is proportion to the current passing through it.
(b) Define and derive coefficient of performance with brief explanation (Diagram is not required).
14. (a) Describe construction and working of Transformer with diagram and derive relative expression.
(b) State Faraday's laws of Electromagnetic induction. Explain Mutual induction and derive an expression for mutual inductance.
15. (a) Draw a labeled diagram of an A.C generator and derive expression for the alternating voltage produced. (Working is not required)
(b) Explain the behavior of Capacitor in AC circuit. Construct the Phasor diagram and derive expression for capacitive reactance.

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ایم ای ایجوکیشن سینٹر

بارہویں: اردو (مڈ ٹرم امتحان: 2025-26)

- (۱۰) سوال نمبر ۱۔ (الف) نظم کے شاعر کا مختصر تعارف تحریر کیجیے۔ (ب) درج ذیل میں سے کسی ایک نظم کا مرکزی خیال تحریر کیجیے۔
- ☆ حضرت فاطمہ الزہراءؑ کی رخصتی ☆ دو بہرے شناساؤں کی ملاقات ☆ بدلی کا چاند
- (۱۰) سوال نمبر ۲۔ (الف) منتخب کردہ سبق کے مصنف کا مختصر تعارف تحریر کیجیے۔ (ب) کسی ایک سبق کا خلاصہ تحریر کیجیے۔
- ☆ سویرے جوکل آنکھ میری کھلی ☆ میرا اہم ☆ سحر ہونے تک
- (۱۵) سوال نمبر ۳۔ مندرجہ ذیل میں سے کسی ایک شاعر کی شاعرانہ خصوصیات تحریر کیجیے۔ (کم از کم دس خصوصیات جن میں نقادوں کی رائے اور حرف آخر بھی شامل ہو)
- ☆ خواجہ میر درد ☆ مومن خان مومن ☆ علامہ ڈاکٹر محمد اقبال
- (۱۵) سوال نمبر ۴۔ مندرجہ ذیل میں سے کسی ایک نثر نگار کے طرزِ تحریر کی نثری خصوصیات تحریر کیجیے۔ (کم از کم دس خصوصیات جن میں نقادوں کی رائے اور حرف آخر شامل ہو)
- ☆ مولوی عبدالحق ☆ سید سلیمان ندوی ☆ علامہ شبلی نعمانی